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Illusions of controlling the future: risk and genetic inheritance

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ABSTRACT *The aim of this paper is to analyse the implications of current beliefs that human beings can control the risk of inheriting a genetic disease and influence their present and future health. To accomplish this goal, materials will be drawn from disparate literatures bearing on concepts of probability, risk and genetic inheritance, and on empirical data gathered on cancer survivors and healthy persons with a family history of cancer. The concept of risk has been theorised on a grand scale but, as Lupton (1999, Introduction, Risk and Sociological Theory, Cambridge University Press) correctly observes, there has been very little empirical work done on how people experience risk as part of their lived world. In this paper, notions of probabilities and risk will be examined as applied to beliefs in genetic inheritance that are shaped by historical and cultural forces and in turn how they shape people's lives. It will be proposed that the belief that knowledge of one's genetic inheritance can control one's future health and disease is an illusion, and also replicates in part a religious notion of predestination.*

Introduction

The aim of this paper is to analyse the implications of current beliefs that human beings can control the risk of inheriting a genetic disorder and influence their present and future health. To accomplish this goal, materials will be drawn from disparate literatures bearing on concepts of probability, risk, and genetic inheritance and on empirical data gathered on women with breast cancer and healthy women with a family history of breast cancer. The concept of risk has been theorised on a grand scale but, as Lupton (1999) correctly observes, there has been very little empirical work done on how people experience risk as part of their lived experience. Here notions of probabilities and risk will be examined as applied to beliefs in genetic inheritance that are shaped by historical and cultural forces and in turn how they shape people's lives. It will be proposed that the belief that knowledge of one's genetic inheritance can control one's future health

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and disease is an illusion, and also replicates in part a religious notion of predestination.

The paper will begin with a brief historical overview of the development of the notion of probability and risk in the West because any discussion of historical developments brings into bold relief the contingent nature of beliefs and practices in a given society, including concepts of probability and risk in the West. Three abbreviated case vignettes will be presented to illustrate the ways in which people's experience of the risk of genetic inheritance of disease, entrenched in what I call the "medicalisation of family and kinship" (Finkler, 2000a, 2001), leads to an illusion of control and, most important, to fear for one's future health. In anticipation of a dynamic debate, the paper will conclude by suggesting that the notion of the risk of genetic inheritance is embedded in Western religious ideologies.

Historical overview

Origins of the concept of probability and risk

A profound change took place in Western thought during the course of the 17th century that challenged traditional authorities of the medieval church and ancient philosophy. With the Scientific Revolution, the new science expounded by men like Bacon gave humans a sense of rational control over their fate. Building on the new scientific rationalism in the 18th century, the Enlightenment thinkers further championed human rationality and freedom from religious authority. But even as the Enlightenment envisioned the control of nature as part of humans' ability to understand the world, the new consciousness led to a new insecurity and a loss of a sense of certainty in a world no longer determined and ordered by God.

Arguably, it may be difficult for modern humans to imagine that the notion of chance and probability was a relatively recent development in Western societies. In actuality, these revolutionary forms of thought, which had their beginnings in the 17th century, introduced the novel notion of probability^a that had not existed in Western thought prior to 1654 (Hacking, 1975). Probability in the context of gambling is ancient, but the concept that random chance conforms to rules of probability had its origins in the 17th century.^b Thomas (1971) noted that during the Scientific Revolution the word "coincidence" as the juxtaposition of unrelated events first made its appearance. A "chance event" was merely one whose causes were not known, but it was now deemed possible to calculate mathematically the probability of an event that involved human beings. In Thomas's words, "it was this nascent statistical sense or awareness of patterns in apparently random behaviour, which was to supersede previous speculation about the causes of good or bad fortune" (Thomas, 1971, p. 656). Rationality could now be reduced to a mathematical calculation and thereby codified to solve practical problems for which solutions were lacking. This innovation contrasted with past notions of rationality based on principles

of certainty (Daston, 1988). The new theories of probability did not seek ultimate truths; rather, the epistemological innovations of the 17th century enabled the gathering of substantial evidence by induction (Shapiro, 1983), which, by calculating random chance occurrences, could be used to support specific hypotheses. This allowed for the recognition that there were degrees of probability as well as degrees of certainty (Shapiro, 1983).

By the 18th century, probability theory found new application in annuities, gambling, insurance (Daston, 1988; Hacking, 1975; Shapiro, 1983), astronomy (Hacking 1975) and in human mortality (Daston, 1988). In the 19th century, chance and probability had become firmly established in Western thought, if not in the consciousness of all individuals, and with it the related notion of normalcy, which held that central tendencies defined human biological and behavioural normalcy, whereas extremes were regarded as abnormal. This new application to statistical information contributed to social control by delimiting the normal and the abnormal based on probability curves (Hacking, 1990, p. 6).

In sum, the theory of probability developed in the 17th century led to the statistical thinking of the 19th century, and soon these theories were applied to medical events. Today, contemporary life continues to be dominated by statistical modes of thought, creating new realities. For example, the human genome, a statistical composite, can be seen to represent one individual (Haraway, 1996) who, in order to act, must now weigh probabilities, as will be seen shortly in the case vignettes. In fact, modern concepts of choice and decision theory, as well as risk analysis are anchored in notions of probability, which are, however, culturally and historically determined.

The concept of risk

Whereas notions of probability had their beginning in the 17th century, the concept of risk became firmly fixed in Western European culture only in the 19th century, although notions of avoiding risk date back as far as the 13th century (Daston, 1988), when the medieval church, which had riled against usury and the taking of interest, had also accepted the right of an individual to be compensated for the risk of time and distance involved in an exchange transaction. Gradually the church accepted risk as a legitimate basis for charging interest on loans (Daston, 1988). However, by the 19th century, the mathematical theory of risk had become widespread and with it the belief that whole classes of phenomena previously taken to be the very model of the unpredictable, from hailstorms to suicides, were in fact governed by statistical regularities (Daston, 1988, p. 164).

At present, the notion of risk control prevails. We now live in, what Beck calls, a "risk society" (Beck, 1992, 1998), where we cannot take "traditional certainties for granted" (Beck, 1998, p. 10). Managing risk is, in his words, "colonizing the future" (Beck, 1998, p. 11) for "events that do not exist (yet) strongly influence our present affairs and activities" (Beck, 1998, p. 11). The notion of

risk assessment makes the “unforeseeable foreseeable or promises to do so” (Beck, 1998, p. 12).

Giddens, along with Beck, emphasises that in high modernity of today, “fate and destiny have no formal part to play” in a world in which humans have “control of the natural and social worlds” (Giddens, 1991, p. 109). The world of “future events is open to be shaped by human intervention [and] is regulated by risk assessment” (Giddens, 1991, p. 109).

Risks exist in all domains of life, including economics, the environment and, most important, in human health. Both Beck and Giddens propose that new risks are manufactured in modern society through modern technology, which produces pollution and numerous other hazards that contemporary human beings must avoid and negotiate at every turn. As every person who is treated by biomedicine or is exposed to the mass media knows, myriad risk factors, including obesity, drinking, smoking, and poor diet, must be avoided to maintain one’s good health. Failure to avoid all such risks will result in the person being blamed for his or her sickness. In this context, it is noteworthy to cite Giddens’s observation that:

medical specialists and other researches produce the materials from which risk profiling is carried out. Yet risk profiles do not remain the special preserve of the experts. *The general population is aware of them, even if it is often only in a rough and ready way.* (Giddens, 1991, p. 120; emphasis added)

According to Rockhill (2001), the idea of risk factors was introduced into health research and brought into health consciousness by Framingham researchers in 1961. “Risk factor logic introduced the notion of probability as a fundamental component of disease causation theories” (Rockhill, 2001, p. 366). Importantly, traditional concepts of risk referred to the statistical profiles for aggregate population. But in today’s collective consciousness, the notion of risk has become privatised (Rockhill, 2001), meaning that it is applied to individuals and not to groups. The prevailing belief now is that the individual is at risk rather than the population of which he or she forms a part. This formulation corresponds to the neo-liberal view of risk that places the responsibility for avoiding and managing it on the individual (Lupton, 1999, p. 5) rather than on external societal factors.

In brief, the concept of risk forecasts the future based on past occurrences. It shifts human fate from the caprices of the gods to the whims of mathematical probability. Paradoxically, the idea of risk, based as it is on mathematical probabilities, is double faced: on the one hand it recognises that events are random, and on the other it believes that this very randomness can be controlled to attain a degree of command of the future. It therefore permits modern humans to nurture the notion that they can dominate the future by controlling risks.

The modern outlook incorporates notions of probability and risk and stands opposed to earlier notions of determinism and fatalism. Giddens (1991) firmly

declared that modernity has rejected fate and determinism. Yet, it is noteworthy that observers of contemporary society propose that modernity and contemporary humans have not fully rejected determinism. For example, Gilkey perceived a profound contradiction in science encompassing determinism and fate^c as well as “a sense of control over the determining fates through intentional activity; that is, rational activity directed at chosen ends” (Gilkey, 1968, p. 66). Science at once encourages humans to seek knowledge freely, to explore nature and human behaviour, while concurrently humans, as understood by science, are also “totally determined beings” (Gilkey, 1968, p. 67). This notion comes into bold relief in Gilbert’s assertion that we must acknowledge that we are “determined in a certain sense, by a finite collection of information that is knowable” (Gilbert, 1992, p. 96). “It is the closing of an intellectual frontier, with which we will have to come to terms” (Gilbert, 1992, p. 96).

The paradox of science that Gilkey recognises is played out in people’s perceptions of the risk of inherited genetic disorders. The ideology of genetic inheritance of disease puts at risk every person with a family history of a disease. By doing so, it leads people to experience both a sense of predestination fostered by genetic determinism, as well as a sense of having control of their health. These contradictions lead to an illusion of both human control and determinism that comes into sharp relief in peoples’ narratives.

Risk and the medicalisation of family and kinship

Inheritance of disease is placed under the concept of a major risk factor, but it is important to repeat that the concept of risk factor refers to populations and not individuals, even though people regard it as a personal risk. Among the numerous risks the individual must confront in present-day life is the risk of inheriting a disease from one’s family, or what I have called elsewhere the “medicalisation of family and kinship” (Finkler, 2000a, 2001). To scholars’ characterisations of modern society as a risk society, we must add that contemporary life is also governed by the “hegemony of the gene” leading to the medicalisation of family and kinship (Finkler, 2000a). By this, it is meant that family and kin relationships are being drawn into the biomedical domain through current understandings that diseases are genetically transmitted from generation to generation and anyone with a family history of a disease is at risk to fall ill sometime in the future.^d Consider this: embedded in concepts of genetic inheritance is the notion that family and kin are the medium through which the inheritance flows. Family and kin connections are framed in terms of genetic inheritance from parents, grandparents, and other relatives.^e As Bobinski observes,

information about genetic makeup is gotten from examination of a person’s medical record and analysis of his or her family genetic history. A family history of a particular disorder, for example, would mean that there was an elevated risk that the individual might be a

carrier who could pass the condition on to a new generation. (Bobinski, 1996, p. 80).

The medicalisation of any human condition impacts dramatically on people's deepest level of experience, understanding and actions, transforming the person from an active being into a passive patient. While the medicalisation process is not a new phenomenon in Western medical history, the medicalisation of family and kinship is relatively recent because the new genetics revolves around it (Finkler, 2000a).

Anglo-American kinship, understood as biogenetic (Schneider, 1980), has come under medical scrutiny through the prevailing explanatory biomedical model of disease aetiology. Our kinship relations have been given a new dimension that stresses faulty genes. In earlier times, and in pre-modern Anglo-American society, people inherited their status, rights and duties, property and power, as well as poverty from their kin. In contemporary times, kinship relations usually do not confer on people any particular status among most of the population.^f In a democratic society, the role of a person's status is denied and/or effaced. In fact, the majority of middle- and lower-class families may have little property to transmit to their descendants other than ephemeral moral values (Finkler, 1994). Instead, cultural significance is given to genetic transmission, for better or for worse. People may take pride in the genetic inheritance of favourable characteristics, but they can also presumably inherit defective genes that transform them into what is described elsewhere as "perpetual patients" (Finkler, 2000a).

The medicalisation of family and kinship may not necessarily affect all people adversely. In fact, it reassures some people that, if no-one in the family, past or present, has suffered from a particular disease, they too, are protected from becoming afflicted. It may furnish the person with a feeling of security, notwithstanding the fact that this sense of protection may be nothing more than an illusion. It also gives meaning to an affliction for which biomedicine lacks an explanation by shedding light on the cause of a senseless disease, and it obviates the necessity to address the questions of "Why me?" and "Why not others?" in the community since the risk of inheriting already predicts the answer. For example, by medicalising family and kinship relationships through concepts of genetic inheritance, the senselessness and randomness of breast cancer can become meaningful by establishing for a woman a connection with her family and ancestors (Finkler, 2000a).

The belief in the risk of acquiring a genetic disease is a two-edged sword. On the one hand, it places the responsibility for the affliction on the individual and the family, which gives people the semblance of control. Again and again, people noted that if only they would avoid related risk factors—for example, by eating well, exercising, refraining from smoking and drinking—all would be well and by so doing they could overcome the inherited risk. Moreover, in the event that a member of the family fell ill, the knowledge that the disease, such as breast or colon cancer, exists in the genetic history of a family, alerts the healthy members

to change their personal habits to avoid falling ill. This knowledge enables people to master the future by acting in the present and avoiding risks.

Conversely, knowing that one is at risk because other members of the family have been afflicted, creates for healthy people a fear of falling ill (Finkler, 2000b) that may even be transmitted to the next generation.⁸ People's reality encompasses a future that is incessantly punctuated by worry. Indeed, the medicalisation of family and kinship alters people's perceptions from, "If I get breast or colon cancer" to "When I get breast or colon cancer", or any one of a hundred diseases believed to be inherited from family and kin. The notion of genetic risk for a disease has now become almost a disease in and of itself (Nelkin, 1992, p. 188). Most important, with the medicalisation of family and kinship, the person's biography, or what was suggested as "life's lesions" (Finkler, 1994), and other extra-somatic factors, including environmental hazards influencing sickness, can be ignored.

What is more, people may no longer need to present complaints to be regarded as sick. They are at risk and therefore they may be considered sick prior to having any symptomatology, other than originating from a family of people with a particular disease. Those people who have a member in the family or kinship group who have suffered from breast cancer, colon cancer, or Alzheimer's disease will be assumed to have inherited it in an autosomal dominant fashion. As Jonsen hypothesises:

Some with monogenic disorders will be patients without symptoms, but sure to have them in the future. Others, with the genetic patterns associated with polygenic or multifactorial disorders, will be known as a schizophrenic or cardiac or cancer patient long before any illness is felt or any pathology damages the organism; indeed, they may never be affected at all, yet still be marked. Persons will become patients before their time: they will be described in disease terms but "feel fine" and "be fine" for years, perhaps always. (Jonsen, 1996, pp. 8–9)

Significantly, too, although risk factors calculated on the basis of probabilities are presented in impersonal and a-moral terms, people interject a moral dimension when they must negotiate notions of randomness and luck. Ironically, the concept of risk gives people an illusion of control, while placing them in a moral bind by holding them personally responsible for its avoidance. Concepts of probability and risk have become equivalent to moral arbiters analogous to religious authorities, custom or community that usually regulated people's behaviour in the past and cross-culturally. People reported that they had done everything correctly and yet they fell ill: they avoided the risk factors for a disease by changing their diets, exercising regularly, ceasing to smoke and drink, but still they fell ill. Some women lamented that they had avoided the risk factors regarded as significant in breast cancer by having their children at a young age, and breast-feeding them, yet they still fell ill. This is a conundrum that leaves people without an explanation and also morally indignant, as becomes apparent in the narratives.

Experiencing risk^h

Excerpted examples from fieldwork data demonstrate these paradoxes as they are played out in peoples' experience of dealing with risk and the ideology of genetic inheritance as a result of the medicalisation of kinship, which gives people at once an illusion of control, as well as, conceptions of predestination. Owing to space limitations, only one abbreviated narrative by a healthy individual and four by women who recovered from breast cancer can be presented in support of the arguments made here.ⁱ

Healthy individual

KK is a 39-year-old college graduate, homemaker, married with two children, whose mother and grandmother had died of breast cancer, and her 77-year-old aunt has developed several types of cancers as well. She observed, "My grandfather's (maternal grandfather) two sisters had breast cancer. So on both sides I get it. And then there was cancer on my father's side."

KK has had mammograms taken regularly since the time her mother fell ill, when KK was 29 years old. Being a devout Christian she claimed that she was totally unconcerned about her genetic heritage because "God protects me", notwithstanding the fact that she followed the doctor's urging to be genetically tested. She claims to not worry about anything she cannot alter. At the time of the interview she was considering a prophylactic mastectomy, despite the fact that she felt protected by God. She said, "I feel like if I had a mastectomy, I think it would be worth it, especially if I could have reconstruction. I would want reconstruction."

KK does not question why her family has been beset by this sickness because, she said,

I feel like God has a plan and I feel that that's the way it should be. I mean, if my father had not died when I was seven I would be a different person. I just feel, its calmness, you know, a taken care of feeling. Faith. It really is very nice.

KK communicates a sense of predestination that may be tied both to the notion of the inevitability of her genetic heritage concomitant with a strong religious faith, which, she claims, protects her from affliction. Yet, behind her assurance of her destiny even in the next world, KK's narrative suggests that she is conflicted by contradictory feelings. This is seen by the fact that, while she accepts her genetic inheritance as her fate, she contemplates possibly having a prophylactic mastectomy done, which would touch at the core of her womanhood, especially because her personhood is tied to the traditional mother and wife roles. KK's religiosity and genetic determinism become co-mingled, yet she considers she can control her future health with prophylactic measures.

Recovered patients

SS, 54 years old and married, immediately told me as I soon as I came into her house: “I have an interesting cancer [case] because I had had prophylactic mastectomies. So I am a person with no breasts who got breast cancer.” SS, referring to her mother, who had died of heart disease at the age of 42 said,

so she might have developed breast cancer [if she had lived]. My grandmother died of breast cancer when she was 70, which metastasised to her liver. Her sister died in her 50s of liver cancer, which we now believe was a metastasis of a breast cancer because she never went to the doctor until she was at death’s door. She would never have done breast exams or anything, so she died of that. My father just died of Parkinson’s last year. I have very, very few people left, but I’ve always been afraid of breast cancer.

SS’s older sister and her cousin had also died of breast cancer, and she has another sister who has breast cancer. SS’s grandmothers’ nieces and great nieces all had breast cancer. After learning about all of the family members who had cancer, SS concluded “there is a strong family history, which was a big surprise to me. I didn’t know all of them had had breast cancer,” because she had not had any contact with them. When SS learned about her family history last year she presumed, “There has got to be some genetic link. And I’m almost positive my great aunt died of this. It was a revelation that this disease had shown up so much in my family.” But, she regrets more, she said “... her fat legs that came from my grandmother than my breast cancer. It’s one of the things you get. You get fat thighs, you get pretty eyes, and you get breast cancer.”

SS never asked herself “Why me?” because she anticipated the affliction in light of her family medical history. In fact, SS observed,

they used to say that it was an injury that brought it [the cancer] about, but of course we know it doesn’t. So I grew up with this almost dread of breast cancer. When I kept having these lumps, I decided to have prophylactic mastectomy. They scooped the breast out and did silicone implants. I remember the surgeon saying “Well you won’t get breast cancer now.”

The doctor was so sure she would not get cancer that he said “‘you don’t have any breasts’. I also do not have a uterus because I had fibroids and had a hysterectomy when I was 40 so this seemed like a no-brainer.” She took oestrogen-replacement therapy for two years and then, “I got this little pea below my chest wall.” Initially, SS ignored it, believing she was totally protected from breast cancer—but it turned out to be breast cancer. She recalled, “It was sitting up on the chest wall. Now they had to take the back muscle because that’s where it was sitting.”

Ironically, SS’s decision to have a prophylactic mastectomy was based both on the notion of predestination, that she will definitely become afflicted by cancer

and on an illusion that she can control her future cancer by having the operation. Interestingly, too, she gives equal weight to her family's entire legacy, her fat legs and her cancer, as if the two legacies were of the same magnitude. What SS failed to recognise is that inheritance of "fat legs" does not necessarily convert the person into a patient.

WW, a born-again Christian, who had breast cancer, immediately informed me:

I did not know what the good Lord had in store for me. The good Lord is just letting me live to see my kids grown and take care of themselves. I've got two grandchildren now. So I have been real blessed. You go through the various states of saying "Why me?", but I had not found the answer, but there's gotta be a reason I'm going through this; maybe it's to touch somebody else's life, give them hope. And the people in the church are always there for you. They are one big family.

When told by her doctors that her disease was hereditary, WW was not surprised because "My grandmother had it, my mother and mother's two sisters had it. I have a cousin that has breast cancer and her cousin's daughter that's 28 just recently found out she had breast cancer." WW's father died of lung cancer. After listing these people, she remembered that she had three brothers, and a sister and a 51-year-old sister who had had both breasts removed even though the cancer was localised in one breast. After identifying the people in her family who had been afflicted with cancer, she concluded, "but I've been fortunate: I've done very well. God's been good to me."

When the doctors had told her that her disease was hereditary they stressed that her girls need "to really stay on top of it". The doctor who performed her reconstructive surgery advised her that after her daughters were married and had their children they may wish to have their breasts and ovaries removed. When a second doctor suggested the same option for her daughters, she mentioned it to the 24- and 18-year-olds because:

I hate to think that I could pass on to them, but it's not anything that any of us planned on. It's just one of those things that happens and if it is, it's just something that you learn to cope with and realise that you know it's not anything that anybody did because no one intentionally wants anyone to get cancer, you know. It gives me more insight that they can do things to help protect themselves better by knowing than not knowing. Making sure they get their check-ups, or not to let anything go unnoticed.

She, in fact, informed her daughters that in all likelihood they too would become afflicted with the disease.

WW believes that knowing that her disease is hereditary is, in her words, "a cautionary measure not that they (her daughters) can help prevent them from going through any of this". But the doctors will keep closer check on them and in this way, she said, the girls will, "just have like a head start that some people

would not have, you know. And I think that's a plus. It bothers me to think that I could pass something on to my children, but like I say, it's not anything that I have done." She finished by declaring a moral indignation when she said:

It's kind of scary to think you could pass something on to someone else. You have no control over that, it's just something that happens and you just deal with it. And you just hope that your children can learn from it and not feel like, you know "she's giving me a death threat".

WW wistfully commented:

I did everything I thought was [right]—I didn't smoke, and I breast fed my last two children. I did things that [made me] less likely to get cancer, breast cancer, but I just didn't luck out. I was just one of those that got it.

WW is worried about her daughters, and also her son, who could pass it on to his daughter. WW worries "that it could pass through generations. I don't want to pass it on but it is nothing that I did wrong."

While WW regards the disease afflicting her family as a matter of predestination, yet, paradoxically, the hereditarian ideology also empowers her to alert her children and their physicians to attend to their health. In this manner, predestination can be controlled, even if her children become "a-symptomatic patients" relegated to constant medical monitoring. Lastly, WW's narrative reveals a modern moral contradiction generated by the notion of risk that confronts her: although she had done everything correctly to avoid all the known risk factors, she still fell ill. Only her church has the potential, and arguably succeeds in resolving this conundrum for her.

Risk, chance, and illusion of control

Humans, as all living creatures, inherit predictable characteristics for their species, but the variation within their species, including disease, is inherited by random chance (Finkler *et al.*, in press; Hadler & Evans, 2001) and shaped by intra-cellular, intra-somatic and external processes (Hubbard & Wald, 1997; Rose, 1997). Sadly, no matter how fervently humans may wish to leave nothing to chance, in the domain of genetic inheritance, chance and randomness govern, be it eye colour or a disease (Hadler & Evans, 2001; Hubbard & Wald, 1997; Rose, 1997). Arguably, most diseases including cancer result largely from mutations induced by external environmental and exogenous factors and/or dietary deficits, rather than from genetic inheritance (Hubbard & Wald, 1997). In the words of the physician Hadler and the geneticist Evans:

In our deterministic fervor, we often forget that whether one develops any disease has less to do with "nature or nurture" than with chance. Random stochastic mechanism is the most important determinant of

disease occurrence and progression. It is a fool's hope [or perhaps blessing] to believe that we will be able to sort out the interactions of environment, 100,000 human genes, and the role of chance to provide a crystal ball that is anything but murky. (Hadler & Evans, 2001, p. 253')

By avoiding identified risk factors, humans may not be able to protect themselves from randomness, or "tame chance" (Hacking, 1990), even though they may have attempted to control their existence from their earliest beginnings by living illusions.

In keeping with this point, it may be fitting to cite Bourdieu's assertion, that all of human existence is lived in terms of what he calls an *illusio* that makes life meaningful. Illusion shields humans from recognising absurdities. In his words:

Only for someone who withdraws from the game completely, who totally breaks the spell, the *illusio* renouncing all the stakes, that is, all the gambles on the future, can the temporal succession be seen as a pure discontinuity and the world appear in the absurdity of a future-less, and therefore senseless, present, like the Surrealists' staircases opening to the void. (Bourdieu, 1980, p. 82, italics in original)

In Rabinow's words, "society could not exist without an ontologically rooted epistemological blindness" (Rabinow, 1996, p. 11) because humans must have the "illusion that they are pursuing something genuinely meaningful in order to act" (Rabinow, 1996, p. 9).

The pursuit of an illusion comes into view with the medicalisation of family and kin when, instead of leading to a sense of fatalism, it becomes reinterpreted into a belief that to comprehend the reason for one's affliction is to prevent it in oneself, one's future offspring or other relatives. Ironically, while the hegemony of the gene, in actuality, diminishes humans as actors because with this belief their health becomes predetermined, the individual is nevertheless moved to action to avoid risk, even though such actions may often prove futile. Nevertheless, knowledge of one's predisposition to or inheritance of a disorder suggests that something can be done to prevent the inevitable, if only by avoiding identified risk factors. The notion of risk and risk factors cloaks uncertainty in degrees and percentages of certainty mediating the uncertainties of living in a post-modern age that Harvey (1989) characterises as one that places the future in chaos because of its uncertainty.

Conclusion

The historical overview, like cross-cultural studies, points to the cultural nature of human beliefs and actions, including Western notions of probability and risk. It must be stressed here, as was done elsewhere (Finkler, 2001), that the truth or falsity of the entire corpus of genetic knowledge developed by Western

science is not being questioned. In the final analysis, **what is most significant is how human beings' experience is influenced by this knowledge.**^k

Traditional beliefs and practices that usually instructed people how to act in given situations have been largely abandoned in contemporary society and replaced by an existence that requires individuals constantly to make choices (Strathern, 1992), one more risky than the next, especially the host of health risks. A person's family medical history presents a formidable risk factor that people must negotiate, and is one of the few in their lives over which they have no choice.

Ideas about probability and risk provide Western humans a means to deal with the uncertainties created by new scientific knowledge that was directed at controlling nature with a degree of certainty (Hacking, 1975) and predict and control the future. Indeed, contemporary humans have harnessed natural forces for many purposes upon which modern technological society is built. But there is a limit to this control because humans cannot determine their genetic inheritance. Our inheritance is neither determined nor controllable at present, although genetic engineering may change this reality in the future. Genetic inheritance is, in the words of one informant, a "crap shoot" ruled by random chance, and no matter how much modern humans may wish to manage their health by controlling this risk (Lewontin *et al.*, 1984; Rose, 1997), they live with an illusion.^l However, the belief in genetic inheritance does promise contemporary humans immortality within the flux of the post-modern world with the inexorable replication of the family's genes from generation to generation, be it disease. The promise of immortality may be yet another reason why notions about genetic inheritance are so easily embraced. The individual existing in a transient world is fastened biologically to the past and future through their genes (Finkler, 2000a). Ironically, in this manner, molecular biology and the genetic model of family and kinship bridge the essentialism of modern science with post-modern experience of uncertainty.

To conclude the discussion, it is fascinating to consider some scholarly debates concerning human efforts to control their existence by means of religion, since to believe that to know one's risk of his or her genetic inheritance as a way of controlling one's future health and well-being is to live with an illusion analogous to that of religious beliefs. According to Freud (1961), the Scientific Revolution, which allowed human beings to control nature, emancipated them from illusions and insecurities. Human beings recognised their helplessness when faced by nature, death and disease and "the less a man knows about the past and present, the more insecure must prove to be his judgment of the future" (Freud, 1961, p 5). Freud rhetorically asked how human beings could protect themselves against the superior powers of nature, of fate that had threatened them. He responded by stating that "religious ideas arose out of need to defend oneself against crushing superior force of nature" (Freud, 1961, p. 4). Humans created gods to exorcise the terror of nature, and reconcile them to the cruelty of fate, death and sickness. However, religions obfuscate reality and they are "not precipitates of experience or end results of thinking: they are illusions,

fulfilment of the oldest, strongest and most urgent wishes of mankind” (Freud, 1961, p. 30). Freud proclaimed that scientific work is “the only road which can lead us to knowledge of reality outside ourselves” (Freud, 1961, p. 31) that is not an illusion. Freud also claimed that religion lost its influence over the human masses with the advent of science.

Arguably, Freud was one of the most influential thinkers of the 20th century, but he was not unique in proclaiming the illusory nature of religion in favour of science. Marx before him also held that religion was the “illusory happiness of men”, and “the opium of the people” (Marx, 1972, p. 12). Among anthropologists, Malinowski (1948) advanced the theory of magic as an illusory form of giving “primitive” humans control of nature and the unknown. Magical rites alleviated a sense of helplessness in the face of unpredictable events.^m Geertz (1973, p. 90), however, attempted to rescue religion from such criticism by regarding religion as draped in factuality and emotional power. For this reason, human beings believe in it and therefore religion was a reality and not simply an illusion.

Contrary to Freud, Gilkey, a contemporary scholar of religion, concludes an incisive discussion of science and religion by stating that:

Today, as throughout history, mankind lives in terms of “myths”—the religious myth that man has been set in the centre of things by a sovereign divine will, or the “scientific” myth that scientific man can sovereignly determine his own destiny and freely alter the march of historical events to suit his chosen moral purposes. Today, as in the past, intelligence requires that we ponder critically the empirical validity of these myths. (Gilkey, 1968, p. 76)

Douglas & Wildavsky (1983, p. 30) would agree with Gilkey that religion has not been displaced by science in the modern world. They do not believe, like Freud, that human propitiation of the gods is but an illusion. On the contrary, for Douglas & Wildavsky propitiation rituals are a means of controlling the future, in much the same way as contemporary humans’ attempt to control future risks. These authors assert that modernisation and science have taken the mystery and awe out of the cosmos and “belittled the gods, or totally discredited them. The same disbelief in religion, which casts such doubt on heavenly intervention in human affairs, gives us the idea that men, not gods can rule the world ...” (Douglas & Wildavsky, 1983, p. 30). Douglas & Wildavsky argue for the universal character of human beliefs concerning dangers, or risks, that are not any different now than in the past.

People have believed before (and boasted) that their technology can deliver all mercies short of immortality. Admittedly, their faith in magic has been misplaced, but that has not prevented them from demanding commissions of inquiry into every accident and postmortems for every death. (Douglas & Wildavsky, 1982, p. 33)

Hence, for Douglas & Wildavsky (1982, p. 47), “primitive” practices, such as

magic, aiming at preventing risk and controlling nature are not illusions, but sound “management” of social realities through cultural concepts of pollution that are “instruments of control”.

While Douglas & Wildavsky (1982) may rightly maintain that humans always knew about risks, which they tried to manipulate by religious propitiation, it cannot be said that there is no difference between propitiating the gods to avert danger and attempting to avoid risks. The gods predetermined the risks in which humans believed and people needed to propitiate them to avoid disasters, whereas now it is believed that humans can control adverse events by avoiding identifiable risk factors. Thus, the dangers human beings had to eschew in the past were well defined, certain, and people employed numerous machinations to avoid them, including food taboos and innumerable rituals. A person could not choose specific risks to avoid, as is expected in contemporary times. At present, humans live with the illusion that it is in their power to prevent sickness by making rational behavioural choices to avoid adverse health outcomes.

Douglas & Wildavsky were correct, however, when they asserted that contemporary humans are not very different from humans of the past at least in one respect: their belief in fate.ⁿ This leads to one other consideration: the notion that, ultimately, scientific beliefs that genetics determines our existence and foreordains one's future health is an illusion that is rooted in the Christian tradition,^o in as much as religious notion of predestination are embedded in the concept of risk of genetic inheritance. If we accept that genetic inheritance is a risk factor, paradoxically, we are taken firmly back to the idea of predestination. Predestination is a concept that emerges out of religious thought that refers to a belief that God predetermines or foreordains whatsoever comes to pass, and in particular the destinies of good and evil (Martin, 1919; Weber, 1958).^p

In his analysis of bioethics, Jonsen (1991) argues persuasively that John Calvin deeply influenced American thought through Puritanism that has become manifest in contemporary theories of bioethics and its basic principles. A similar argument can be made with respect to beliefs in genetic inheritance. Although most scientists would deny that they hold a religious perspective (Pollack, 2000), in reality, the Human Genome Project, now the chief promoter of the new genetics, is under-girded by religious ideas, if not uniquely by Puritanism.^q The idea of genetic determinism^r shares various characteristics with religious ideologies, especially as discussed by some of the proponents of the human genome. The human genome is often described in terms of the “Book of Life”^s and the “Holy Grail”. In her original analysis of the history of the genetic code, Kay observes that “scriptural representations of the genome as the Book of Life have come to signify a kind of divine intervention” (Kay, 2001, p. 297). References to the Holy Grail made by various scholars, for example Gilbert (1992) and Kevles (1992) in their discussions of the genome are particularly revealing.^t While the Grail pertains to Christ's Cup that contains the blood that resulted from His wounds, frequently it refers to “the idea of the quest for the secret of life” (Ross & Ross 1919, p. 386).^u

Francis Collins (1999), head of the Human Genome Project, stated that he

sees no conflict between religion and genetics and that uncovering the genome is a “religious experience”. He also wrote,

if healing is something which Jesus Christ in his short time on this earth spent so much time on, it is something that those who would follow him should also consider especially important. Such is the theological justification for the Human Genome Project. (Collins, 1997, p. 95)

Collins observed that “understanding the genes that play a predictive role in diseases will facilitate the development of new treatments. It is part of our mandate as Christians to pursue such medical advances, attempting to emulate Christ in his healing role” (Collins, 1997, p. 96).

These considerations bring us back to Freud’s claim that the scientific enterprise would subvert religion. Yet, religion flourishes in contemporary America even though many theologians recognise that the power of humans to shape their “own future is concentrated in the biological sciences” (Hefner, 1968, p. 6). If we accept the notion that medicine and related fields are embedded in Western culture and values—and there is an extensive literature to support this proposition^v—then we need not be surprised that the belief in genetic inheritance echoes a religious ideology of predestination. Religious ideology has been naturalised to the extent that the Human Genome Project unreflectively and seamlessly slides into the taken-for-granted world and theories of risk are experienced as morally right. For this reason, perhaps it can even be said that religion is no longer a significant factor in modern humans’ existence and that scientists tend not to recognise that the ideology of genetic determinism is embedded in religion with its illusions. The modern humans’ illusion is that we have become emancipated from illusions.

Contemporary people’s belief in genetic inheritance shields people from confronting the chaos of events with illusions in much the same way as earlier notions embedded in religion had done. Ultimately, then, contemporary humans’ ideology of control of the future through controlling the risk of inheriting a genetic disease is a comforting necessity, especially in an age of uncertainty and flux.

As the ideology of genetic inheritance continues to penetrate the fabric of post-modern society cross culturally, it will be necessary to examine how people in non-Western societies negotiate the notion of risk and risk factors, especially as moral imperatives, given that these notions are embedded in Western history and Western thought, and how people worldwide reinterpret their religious beliefs to fit the cloak of secularism.

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Notes

- (a) Following Hacking, probability refers both to a tendency displayed by some chance occurrences to produce stable relative frequencies and a degree of belief that is supported by evidence from nature and material things rather than by people's testimonies (Hacking, 1975). Additionally, according to Daston, "mathematical probability theory was to be the codification of a new brand of rationality that emerged approximately the same time as the theory itself, or rather of a more modest reasonableness that solved everyday dilemmas on the basis of incomplete knowledge, in contrast to the traditional rationality of demonstrative certainty" (Daston, 1988, p. xi).
- (b) While prior to the 17th century ideas about determinism prevailed, human beings attempted to control and even predict the future prior to the development of the laws of probability by acts of randomisation; for example, when fowls were cut up, or by practices such as scapulamancy (Moore, 1957). Humans used divination throughout history to help them make decisions about matters for which they lacked information (Wallace, 1966). Divination supplied people with the missing information by "direct, if apparently arbitrary, advice from supernatural authority" (Wallace, 1966, p. 108). Azande oracles, as described by Evans Pritchard (1937), weighted probabilities by consulting a poison oracle, which instructed the person on which actions to take. In the ancient art of astrology, which was based on a study of movements of the heavenly bodies, the astrologer could "construct a horoscope for some future point of time, and thus predict the influence the heavens would exert on that occasion" (Thomas, 1971, p. 285). But the astrologers were always at the mercy of the heavens, which ultimately controlled future events (Thomas, 1971).
- (c) Gilkey defines fate as "blind forces of nature, of genetic inheritance, of disease and uncontrolled scarcity of social structure and tradition of local or family custom and most important, of the inner psyche of all those forces that shape human existence in directions antithetical to or destructive of conscious human goals and purposes" (1968, p. 66–67).
- (d) On the program *Sixty Minutes II*, Charlie Rose interviewed Francis Collins, the head of the Human Genome Project, who predicts that, in the near future, genes for all diseases or for the disposition to all diseases, including anorexia nervosa and cardiovascular diseases, will be identified on the human genome (*Sixty Minutes II*, 2002) and all diseases will become controllable through genetics.
- (e) *Science* (1993) describes how the BRC-1 gene for breast cancer was discovered by dependence on blood samples and family histories from numerous extended families.
- (f) Except, arguably, among elites in America (see Bellah *et al.*, 1985; Marcus & Hall, 1992).
- (g) This became especially clear to me in a recent as yet unpublished study of colon cancer patients and healthy individuals with a family history of colon cancer. One of the subjects and his wife reported that his young daughter lived with great fear that she has inherited the disease from her father and will similarly become afflicted.
- (h) A multi-sided ethnographic approach was used and in depth interviews were done with cancer survivors and individuals with a family history of cancer. A few questions were posed to the participants in the study, the main one being "What does it mean to experience breast cancer or come from a family with history of cancer"?
- (i) For extensive narratives by 35 people, further analysis and a discussion of the methodology, see Finkler (2000a).
- (j) See also Dawkins (1989), one of the most prominent spokesmen for genetic determinist; and also note (r).

- (k) Writing about post-traumatic stress a similar point is made by Young (1995), who argues correctly that it is not for him to deny or affirm the reality of this phenomenon but to examine the ways in which ideas of post-traumatic stress enter peoples' lives.
- (l) Arguably, only genetic engineering may give some actual control of one's genetic make-up and that is still far off in the future.
- (m) According to Malinowski (1948), the difference between magic and religion is that magical rites are directed towards very specific and practical ends, whereas religious ceremonies are performed with no specific purpose but are directed to future events. In both instances, humans recognise that they are at the mercy of higher forces. Malinowski's view is that magic is a rudimentary form of science because like science it contains a "body of rules and conceptions, based on experience and derived from it by logical inference, embodied in material achievement and in a fixed form of tradition and carried on by some sort of social organization" (Malinowski, 1948, p. 17).
- (n) Undoubtedly there are numerous similarities between contemporary and traditional societies, but my current concern is only with the notion of predestination.
- (o) This point is developed in great depth elsewhere (Finkler, 2002, unpublished data). Here the suggestion is made for the readers' consideration.
- (p) See especially Weber (1958) for his analysis of Calvinist ethic in the context of notions of predestination.
- (q) Jones (1996) argues that Calvin was the ultimate genetic determinist. In fact, he identifies several similarities between genetics and religion, especially notions of predestination and original sin and he rejects both. Unlike Jones (1996), who demonstrates that theology and genetics pose similar questions, my argument is that ideology of genetics is buttressed by theological issues.
- (r) This is especially prevalent in evolutionary psychology, one of its advocates being Richard Dawkins, who claims that "we are survival machines, robot vehicles blindly programmed to preserve the selfish molecules known as genes" (Dawkins, quoted in Yancey, 1998).
- (s) The Book of Life first appears in Jewish scripture in Psalms 69 verse 29, when it is said that those who lack merit should be "blotted out of the book of the living". The idea is further developed in the New Year's (Rosh Hashanah) tractate 16a, which refers to divine judgement. For a discussion of the Book of Life in the Christian tradition see Kay (2001).
- (t) Nelkin (1996) is one of very few scholars who has observed the affinity between religion and modern genetics.
- (u) The quest for the Holy Grail represents various significations, including a search for spiritual enlightenment (Ross & Ross, 1919; Waite, 1961; Weston, 1965). Additionally, the Grail "became the emblem of moral purity or of triumphant faith" (Ross & Ross, 1919, p. 386). Gruenwald (1997) believes that the Holy Grail is a search for transcendence, a struggle between good and evil, and a quest for meaning that is especially relevant in modern times marked by human confrontations with the "crisis of credibility", owing to a loss of "certainty of our [religious] conviction" (Holm, 1994, p. 26).
- (v) For an extensive bibliography that explores this assertion see Finkler (2000a). One of the earliest proponents of this view was Fleck (1979 [1935]).

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